

Minor- II: Real Analysis

College For Integrated Studies
University of Hyderabad

March 30, 2026

Duration: 60 minutes
Maximum Score: 20 points

Instructor: Dharmendra Kumar
Course Code: MM 263

Instructions: No mark will be given for writing only TRUE or FALSE (without justification).

1. TRUE or FALSE: Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p \leq 1$. [5]

2. TRUE or FALSE: Series $\sum_{n=1}^{\infty} \frac{(n+3) 5^n}{(2n+1) 3^n}$ is convergent. [5]

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$
TRUE or FALSE: f is continuous at 0. [5]

4. TRUE or FALSE: Every continuous function $f : [0, 1] \rightarrow \mathbb{R}$ is bounded. [5]

All the best !